usage 1 Programming Techniques (e.g. time, memory and storage space). For maximum efficiency we wish to minimize resource usage. However, the various resources cannot be compared directly, so which of two algorithms is considered to be more efficient often depends on which measure of efficiency is considered the most important, e.g. the requirement for high speed, minimum memory usage is considered.

There may be several algorithms for solving a particular problem. How fast a problem can be solved, depends on efficient algorithm. Therefore, analysis of algorithms is very important to compare algorithms and conclude that which one is better than the other. Analyzing the efficiency of algorithms means comparing efficiency of different methods of solutions of a problem.

1.2.4 ALGORITHMS FOR SOLVING PROBLEMS

This section presents algorithms to solve various mathematical problems. This develops skills in finding solutions of problems.

Problem 1: Find the sum, product and average of five given numbers.

Planning the Solution:

Input: Five given numbers

Required Output: Sum, product and average of five numbers

Processing: Addition, multiplication and division of numbers

Algorithm:

Step 1: Start

Let the five numbers be A=2, B=5, C=8, D=4 and E=12

Step 2: FIND the sum (SUM)

SUM=A+B+C+D+E

Step 3: FIND the product (PROD)

PROD=A*B*C*D*E

Step 4: FIND the average (AVG)

AVG=SUM/5

Step 5: Output SUM, PROD, AVG

Step 6: Stop

Problem 2: Find the largest of three unequal numbers.

Planning the Solution:

Input: Three unequal numbers

Required Output: The largest of three numbers

Processing: Comparison of each number with the other two one by one **Algorithm:**

Step 1: Start

Let the three numbers be A=10, B=20 and C=30

Step 2: Check if A is the largest number

IF A>B and A>C THEN LARGEST=A otherwise GOTO Step 4
Step 3: GOTO Step 7

Let the height, H be 10 and the base, B be 15
Step 2: CALCULATE the area, A
A = (B*H)/2
Step 3: Output A
Step 4: Stop

Problem 6: Read marks (M) and print letter grade according to the following scheme.

Marks	Letter Grade
M≥80 and M≤100	A1
M≥70 and M<80	A
M≥60 and M<70	B
M≥50 and M<60	C
M≥40 and M<50	D
M≥33 and M<40	E
M≥0 and M<33	Ungraded

Planning the Solution:

Input: Marks
Required Output: Letter Grade
Processing: Check in which range the marks fall and accordingly print the letter grade.

Algorithm:

Step 1: Start
Let the marks, be M
Step 2: Compare the marks
IF M≥80 THEN print “Grade A1” otherwise GOTO Step 4
Step 3: GOTO Step 13
Step 4: Compare the marks
IF M≥70 THEN print “Grade A” otherwise GOTO Step 6
Step 5: GOTO Step 13
Step 6: Compare the marks
IF M≥60 THEN print “Grade B” otherwise GOTO Step 8
Step 7: GOTO Step 13
Step 8: Compare the marks
IF M≥50 THEN print “Grade C” otherwise GOTO Step 10
Step 9: GOTO Step 13

Step 4: Check if B is the largest number
IF B>A and B>C THEN LARGEST=B otherwise GOTO Step 6
Step 5: GOTO Step 7
Step 6: LARGEST=C
Step 7: Output LARGEST
Step 8: Stop

Problem 3: Find acceleration of a moving object for given mass and the force applied.

Planning the Solution:

Input: Mass and the Force
Required Output: Acceleration of moving object
Processing: Divide the force by the mass

Algorithm:

Step 1: Start
Let the mass (M) be 50 and the force (F) be 12
Step 2: CALCULATE the acceleration (A)
A=F/M
Step 3: Output A
Step 4: Stop

Problem 4: Find the volume of a cube.

Planning the Solution:

Input: Length of side of cube
Required Output: Volume of cube
Processing: Side of cube raised to the power of three

Algorithm:

Step 1: Start
Let the length of one side of cube, L be 12
Step 2: CALCULATE the volume, V
$V=L^3$
Step 3: Output V
Step 4: Stop

Problem 5: Find the area of a triangle when the lengths of height and base are given.

Planning the Solution:

Input: Height and base of triangle
Required Output: Area of triangle
Processing: Multiply height with base and divide by two

Algorithm:

Step 1: Start

Step 12: Compare the marks

IF $M \geq 33$ THEN print "Grade E" otherwise print "Ungraded"

Step 13: Stop

Problem 7: Find the exponent of a given number.

Exponent or power of a number means how many times to use the number in a multiplication. In other words it is the product of a number that is multiplied as many times as its exponent.

Planning the Solution:

Input: A number and its exponent

Required Output: Exponent of given number

Processing: Multiply the number as many times as its exponent

Algorithm:

Step 1: Start

Let the number, N be 8 and its exponent, E be 5

Step 2: Initialize product(P) and K to 1

P=1, K=1

Step 3: FIND the product(P)

P=P*N

Step 4: Increment K by 1

K=K+1

Step 5: Check if the value of K is less than or equal to E

IF $K \leq E$ THEN GOTO Step 3 otherwise GOTO Step 6

Step 6: Output P

Step 7: Stop

Problem 8: Print odd numbers from 1 to 100.

1 3 5 7 9 11 . . . 99

Planning the Solution:

Input: This problem has no input

Required Output: Printing odd numbers from 1 to 100

Processing: Initialize a variable to 1 and keep printing it with an increment of 2 till 99

Algorithm:

Step 1: Start

Initialize variable K to 1

K=1

Step 2: Output K

Step 3: Increment K by 2

K=K+2

Step 4: Check if the value of K is less than 100

IF K<100 THEN GOTO Step 2 otherwise GOTO Step 5

Step 5: Stop

Problem 9: Print the following sequence of numbers in descending order.

27 24 21 18 15 12 9 6 3 0 -3 -6

Planning the Solution:

Input: This problem has no input

Required Output: Printing numbers from 27 to -6 in descending order with a step of -3

Processing: Initialize a variable to 27 and then keep printing it with a decrement of 3 till -6

Algorithm:

Step 1: Start

Initialize variable K to 27

K=27

Step 2: Output K

Step 3: Decrement K by 3

K=K-3

Step 4: Check if the value of K is greater than or equal to -6

IF K≥-6 THEN GOTO Step 2 otherwise GOTO Step 5

Step 5: Stop

Problem 10: Find the sum of even numbers up to 100.

SUM = 2 + 4 + 6 + 8 + 10 + 12 + 14 + . . . + 100

Planning the Solution:

Input: This problem has no input

Required Output: Sum of even numbers up to 100

Processing: Add all the even numbers from 2 to 100

Algorithm:

Step 1: Start

Initialize SUM to 0 and K to 2

SUM=0, K=2

Step 2: ADD K to SUM

SUM=SUM+K

Step 3: Increment K by 2

K=K+2

Step 4: Check if the value of K is less than or equal to 100

IF $K \leq 100$ THEN GOTO Step 2 otherwise GOTO Step 5

Step 5: Output SUM

Step 6: Stop

Problem 11: Print a multiplication table of a given number.

Planning the Solution:

Input: Number whose table is required

Required Output: Printing table of given number

Processing: Initialize a variable to 1 and print its product with the given number. Increment the variable by 1 and print the product. Continue the process till printing the product of 12 and the given number.

Algorithm:

Step 1: Start

Let the number, N be 7

Step 2: Initialize K to 1

K=1

Step 3: FIND the product

P=N*K

Step 4: Output N, K, P

Step 5: Increment K by 1

K=K+1

Step 6: Check the value of K

IF $K \leq 12$ THEN GOTO Step 3 otherwise GOTO Step 7

Step 7: Stop

Problem 12: Convert temperature from Fahrenheit to Celsius.

Planning the Solution:

Input: Temperature in Fahrenheit

Required Output: Temperature in Celsius

Processing: Compute the temperature in Celsius from Fahrenheit using conversion formula

Algorithm:

Step 1: Start

Let the temperature in Fahrenheit, F be 100

Step 2: CALCULATE temperature in Celsius (C)

C = 5/9(F-32)

Step 3: Output C

Step 4: Stop

Problem 13: Find factorial of a given number.

Planning the Solution:

Input: Given Number

Required Output: Factorial of given number

Processing: Find the product of all the numbers from 1 to the given number. **Algorithm:**

Step 1: Start

Let the number, N be 5

Step 2: Initialize loop variable K and factorial F to 1

K=1, F=1

Step 3: CALCULATE the product

F=F*K

Step 4: Increment K by 1

K=K+1

Step 5: Check the value of K

IF K≤N THEN GOTO Step 3 otherwise GOTO Step 6

Step 6: Output F

Step 7: Stop

1.3 FLOWCHART

1.3.1 THE FLOWCHART

Flowchart is a diagrammatic representation of algorithm. It describes what operations are required to solve a given problem.

1.3.2 IMPORTANCE OF FLOWCHART IN SOLVING A PROBLEM

Flowchart illustrates the sequence of operations to be performed to solve a problem in the form of a diagram. Computer programmers draw flowcharts before writing computer programs. It provides an easy way to analyze and find solutions of problems. Once, the flowchart is drawn, it becomes very easy to write the program in any high level language. It is very helpful in communicating the problem solving method to other people. It also helps in finding and removing errors in computer programs.

1.3.3 STEPS FOR DRAWING FLOWCHART

The flowchart developer must determine the following requirements for the given problem or algorithm before drawing a flowchart.

- Input to the flowchart
- Type of processing required
- Decisions to be taken

- The output to be produced after processing

Input: The flowchart developer must know what exactly the input to the flowchart is. The input is determined from the problem statement. For example, the given problem is to convert temperature from Fahrenheit to Celsius. Here, the input will be the temperature in Fahrenheit.

Processing: The flowchart developer must decide what type of calculation is to be performed or which formula is to be applied to obtain the required result. For example, to find the area of a triangle, the following formula is to be used.

Symbol Description	Symbol Shape	Symbol Description	Symbol Shape	
Flow Line		Process		
Start/Stop		Decision		
Input/Output		Connector	On-Page	Off-Page

$Area = (Base * Height)/2$

Decision: The flowchart developer must decide which control structures (sequence, repetition or selection) are to be applied for the solution of the problem. For example, selection structure must be applied to print letter grade of a student based on the marks obtained. The selection structure will check in which range the marks fall and accordingly print the grade.

Output: The flowchart must provide the required output.

1.3.4 FLOWCHART SYMBOLS

Flowcharts are drawn using standard symbols. These symbols have specific meaning and are connected by arrows indicating the flow from one step to another. Flowchart symbols are illustrated in Table 1-1.

Table 1-1 Flowchart symbols

Flow Line: It is a line with arrow head used to connect various flowchart symbols and indicates the flow of control in the flowchart.

Start/Stop Symbol: It is a rounded rectangular shaped symbol. It is used to indicate the start or end of a flowchart. We can only write the words Start or Stop inside this symbol. A flowchart can only have one start but it may have many ends.

Input/Output Symbol: Parallelogram represents input or output operations in a flowchart. It contains the word READ or INPUT along with the variables for input operation or PRINT or OUTPUT along with the output data for output operation.

Process Symbol: A rectangular block is used for any data processing operation. All the

calculations appear inside the processing symbol, such as "SUM= A + B". Variables are also initialized inside the process symbol, such as “K=1”.

Decision Symbol: A diamond shaped symbol represents decision in a flowchart and it contains a condition. If the condition is true, the path marked YES is to be followed. If the condition is false, the path marked NO is to be followed. The words TRUE or FALSE can also be used instead of YES or NO.

Connector Symbols: These symbols are used to connect one part of a flowchart to the other on the same page (On-Page connector) or on the new page (Off-Page connector).

1.3.5 FLOWCHARTS TO SOLVE PROBLEMS

In this section algorithms presented in the previous section will be converted into flowcharts for better understanding of the logic of problem solving.

Flowchart 1: Flowchart to find the sum, product and average of five numbers

Before drawing the flowchart to solve this problem, the input to the flowchart, type of calculation to be performed and the required output must be defined. From the statement of the flowchart, it is clear that sum, product and average are to be calculated and given are five numbers.

Flowchart to find sum, product and average of five numbers is shown in Fig.1-1.

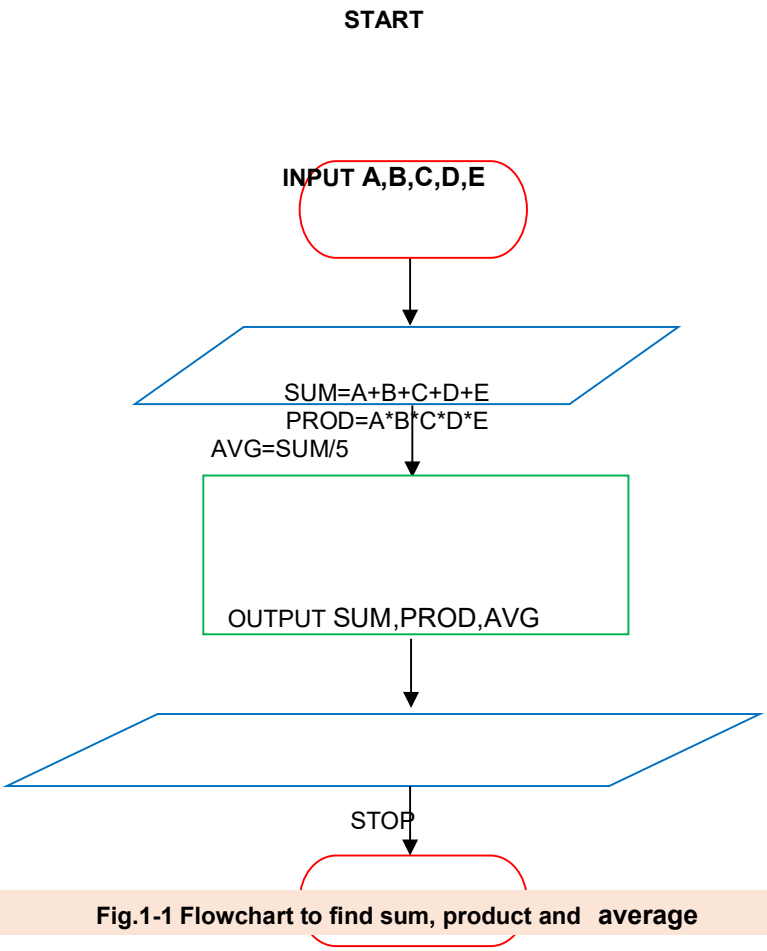

Fig.1-1 Flowchart to find sum, product and average

Flowchart 2: Flowchart to find the largest of three unequal numbers

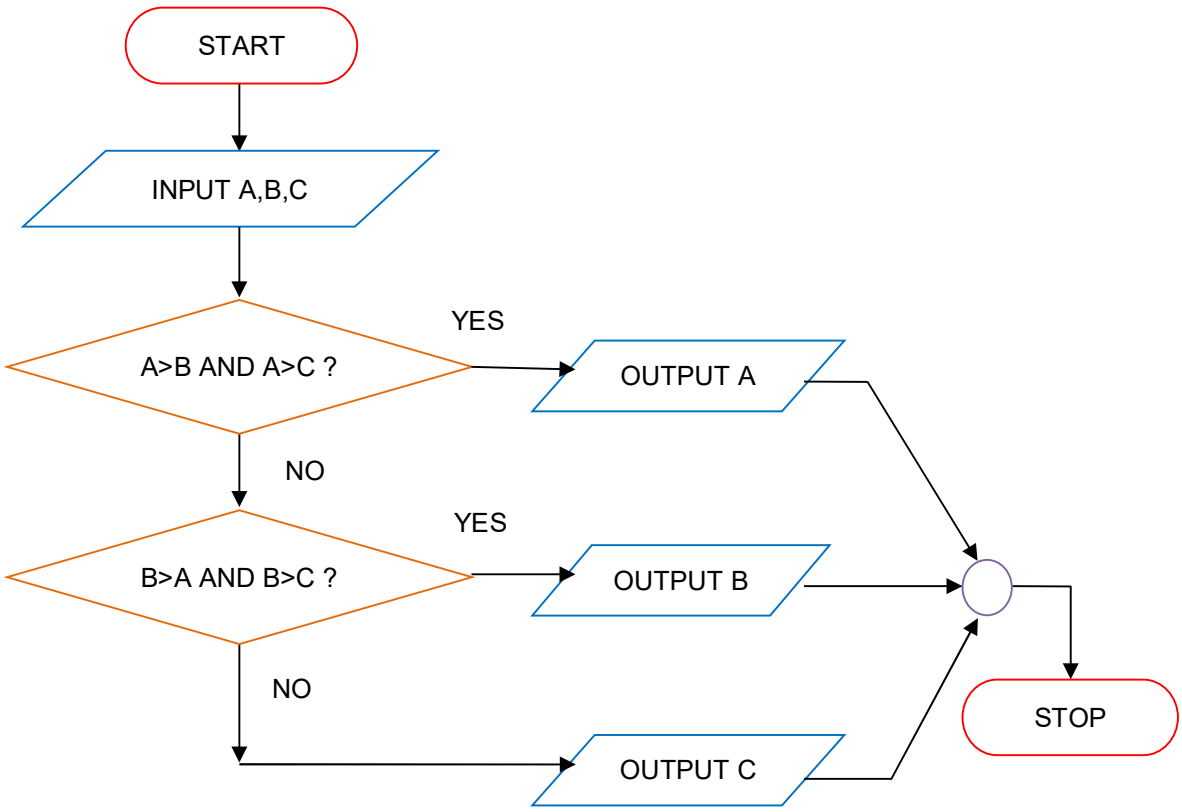

Fig.1-2 Flowchart to find largest of three numbers

Flowchart 3: Flowchart to find acceleration of a moving object given mass and the force applied

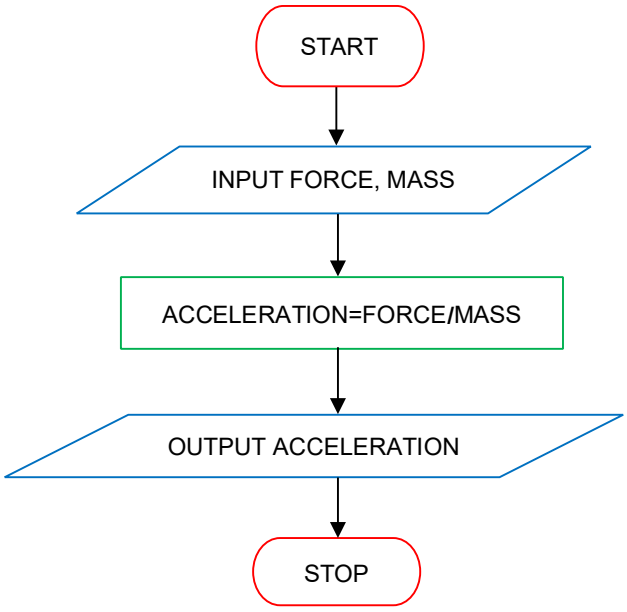

Fig.1-3 Flowchart to find acceleration

Flowchart 4: Flowchart to input the length of one side of cube and print its volume

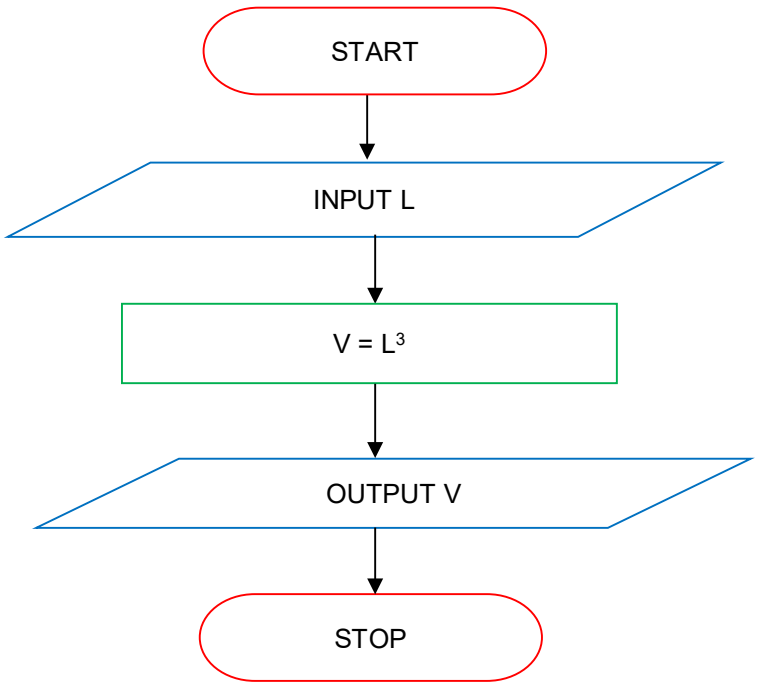

Fig.1-4 Flowchart to find volume of a cube

Flowchart 5: Flowchart to find the area of a triangle when the lengths of height and base are given

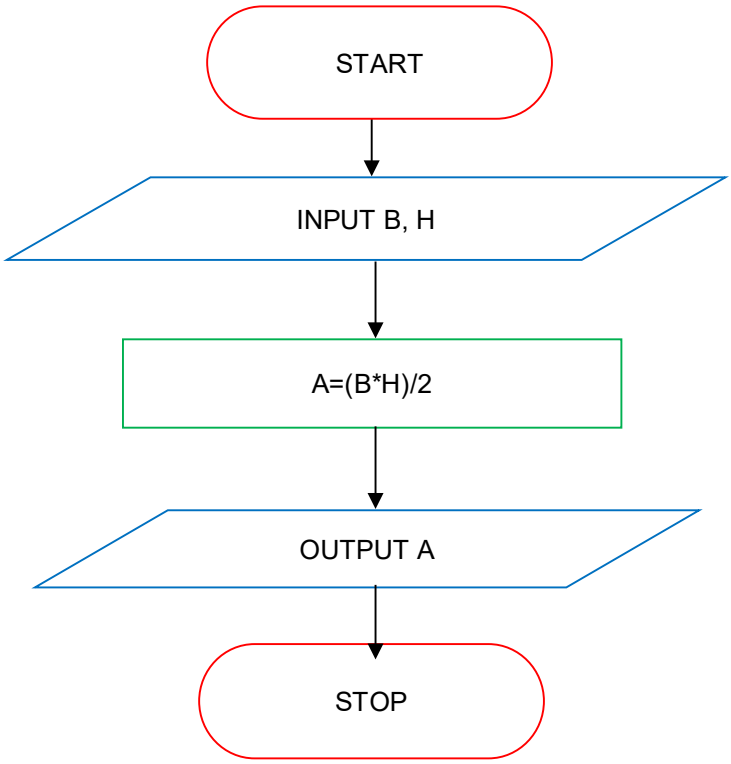

Fig.1-5 Flowchart to find area of a triangle

Flowchart 6: Flowchart to read marks (M) and print letter grade according to the following criteria.

Marks	Letter Grade
M≥80 and M≤100	A1
M≥70 and M<80	A
M≥60 and M<70	B
M≥50 and M<60	C
M≥40 and M<50	D
M≥33 and M<40	E
M<33	Ungraded

Flowchart to print letter grade is shown in Fig. 1-6.

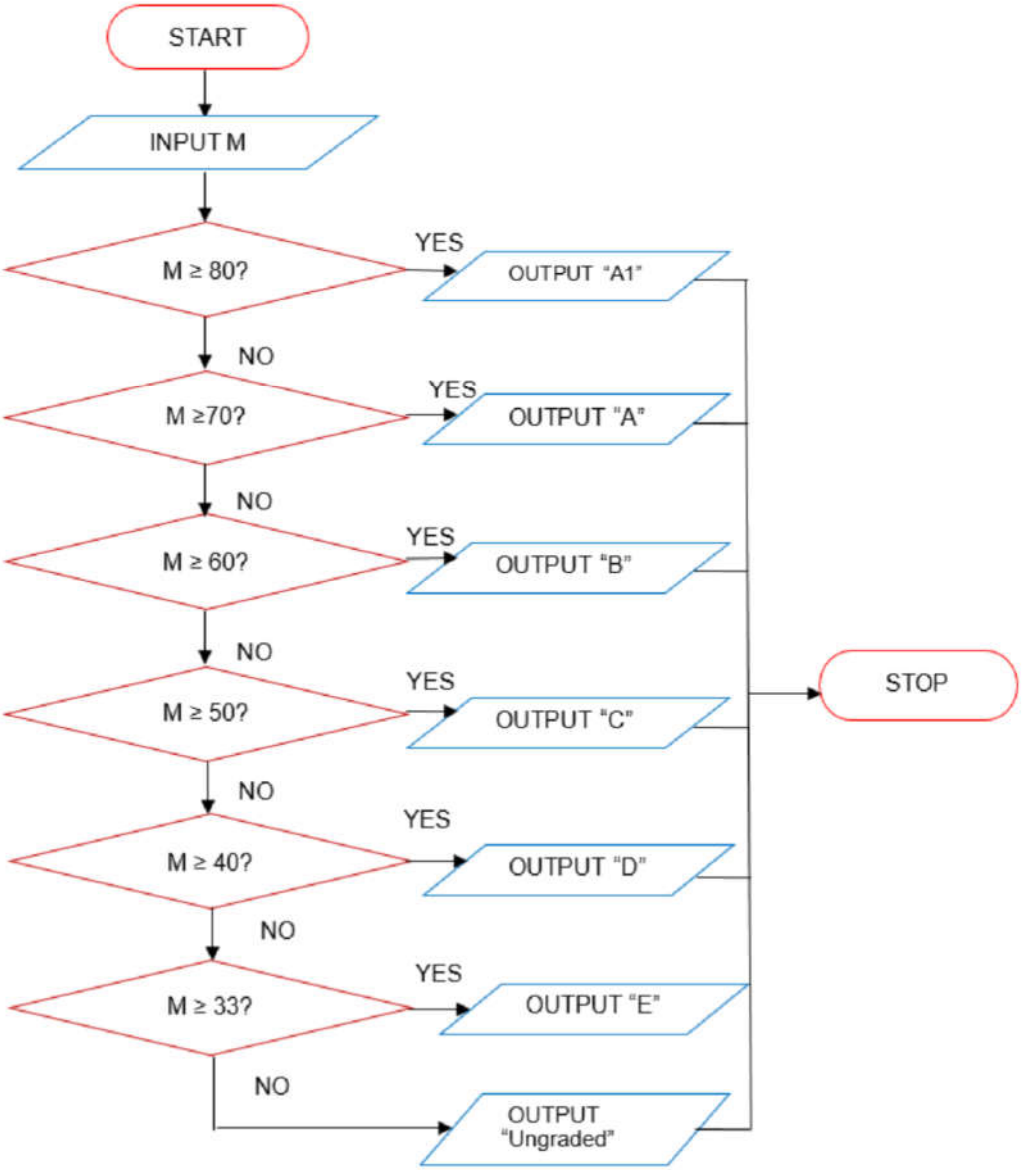

Fig.1-6 Flowchart to print letter grade

Flowchart 7: Flowchart to find the exponent (E) of a given number (N)

In this flowchart, the value P represents N raised to the power E.

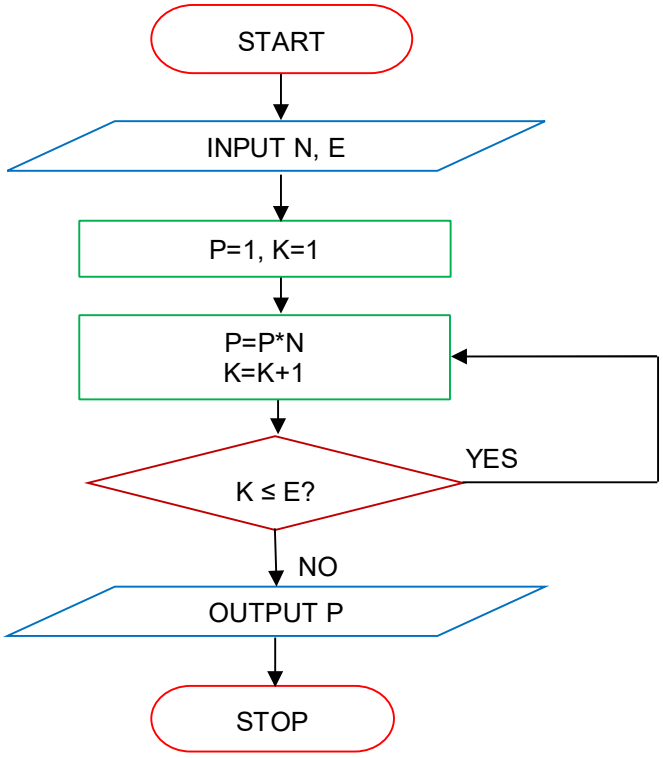

Fig.1-8 Flowchart to find exponent of a number

Flowchart 8: Flowchart to print odd numbers from 1 to 100

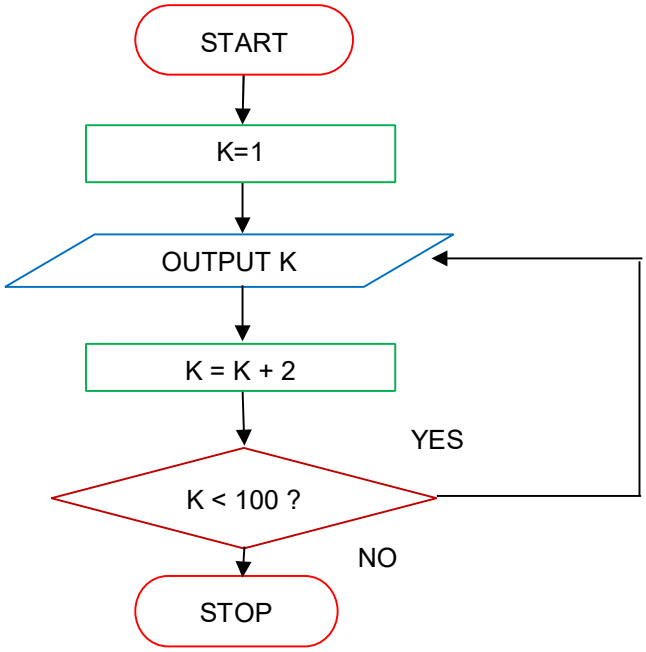

Fig.1-9 Flowchart to print odd numbers

Flowchart 9: Flowchart to print the given sequence of numbers in descending order 27 24 21 18 15 12 9 6 3 0 -3 -6

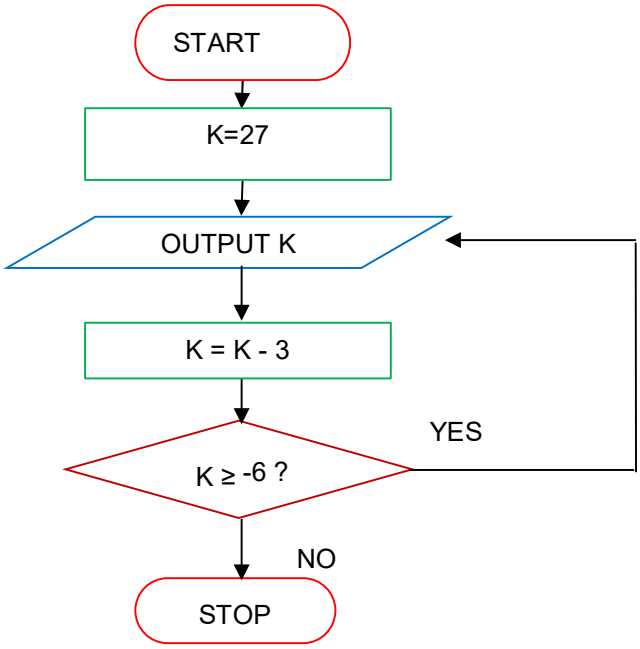

Fig.1-10 Flowchart to print numbers in descending order

Flowchart 10: Flowchart to find the sum of even numbers up to 100

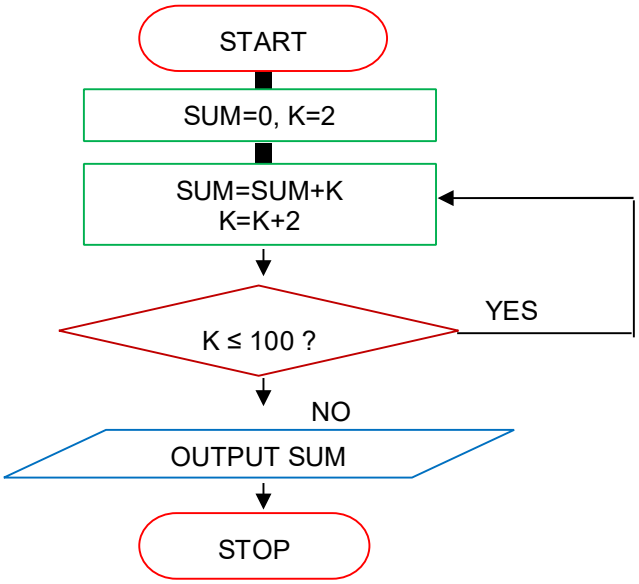

Fig.1-11 Flowchart to find the sum of even numbers

Flowchart 11: Flowchart to print a multiplication table of a given number

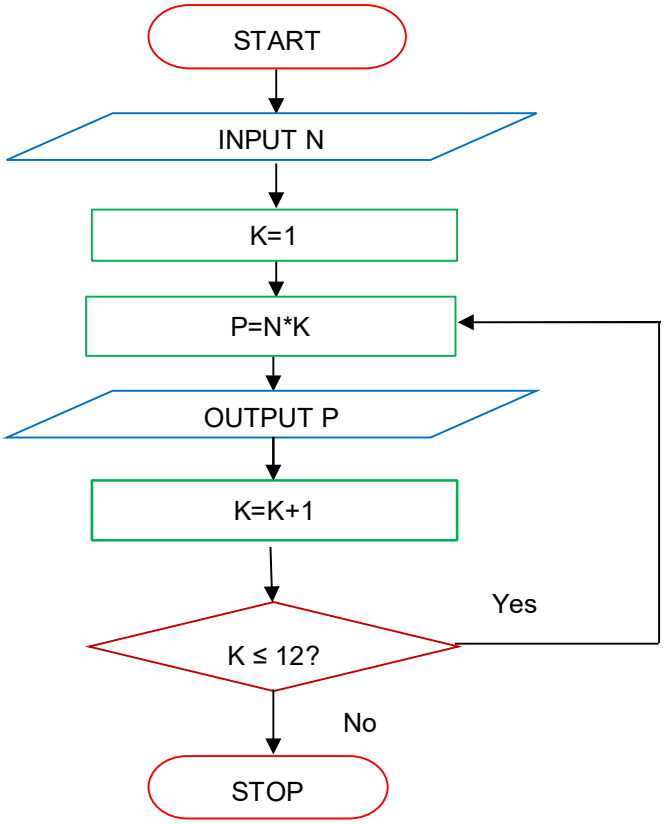

Fig.1-12 Flowchart to print multiplication table of a number

Flowchart 12: Flowchart to convert temperature from Fahrenheit to Celsius

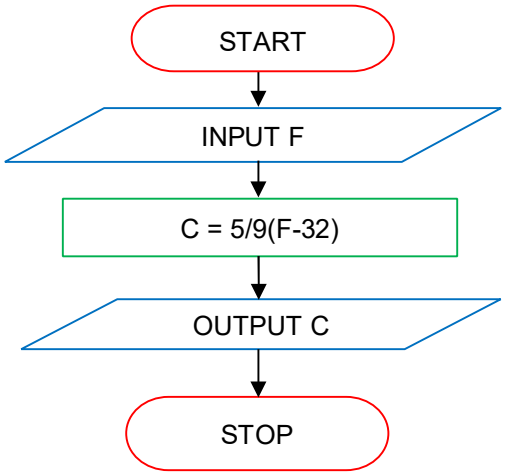

Fig.1-13 Flowchart to convert temperature

Flowchart 13: Flowchart to find factorial of a number

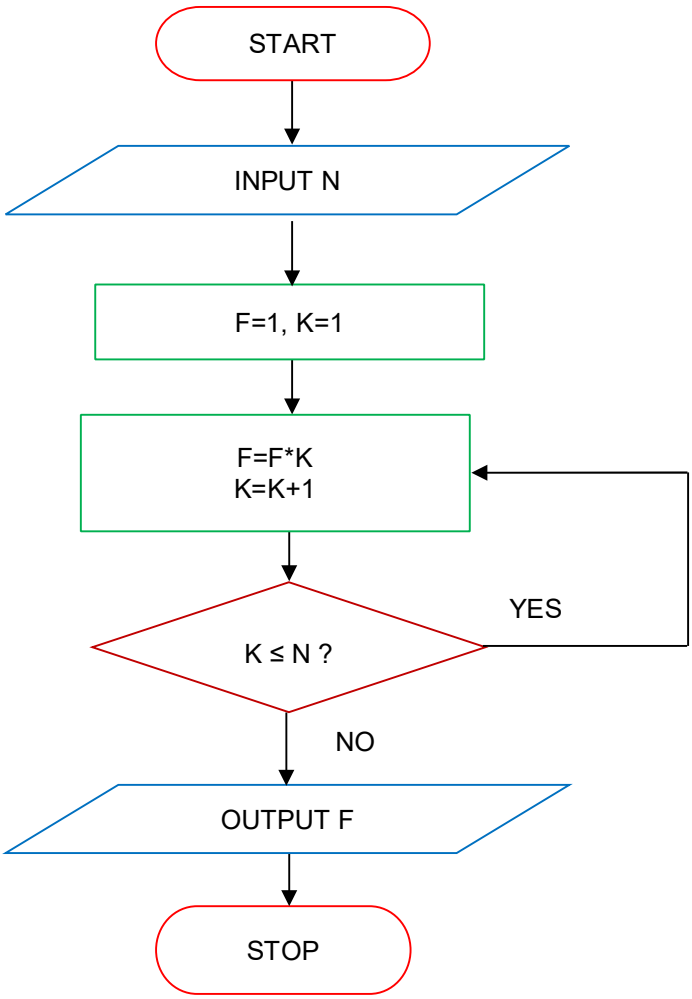

Fig.1-14 Flowchart to print factorial of a number

Key Points

- **Computer** is a general- purpose electronic machine invented to help people solve various problems.
- **Candid solutions** of a problem are all the possible solutions for a problem.
- **Algorithm** is a step by step problem solving method that is easy to understand and follow. It is a set of steps that clearly defines a sequence of operations to solve a problem.
- **Flowchart** is a diagrammatic representation of algorithm.
- **Sequential Structure** refers to the execution of operations in the order in which they appear.
- **Repetition Structure** or **loop** provides a way to repeat one or more operations as many times as required to solve a problem.
- **Selection Structure** allows a choice among various options while solving a problem